Asynchronous Circuits – Part 1
K. Sridharan

Review

- We have discussed combinational circuits, sequential logic circuits and state machines.
- We will now briefly discuss asynchronous circuits.

Note

- Asynchronous circuits are not described in detail in the prescribed text (Mano-Ciletti).
- So we will only touch upon a few things here.

What have we discussed so far

- In the state machines topic in particular, we looked at synchronous sequential logic circuits where a clock signal was used to synchronize the operation.

What is an asynchronous circuit ?

- An asynchronous circuit is one where the operation is not synchronized by a common clock or other means.

Have we seen any asynchronous circuits at all ?

- Question for students.

Give examples of asynchronous circuits

- Question for students.

Example of an Asynchronous Circuit

- The ripple counter we have seen is an asynchronous circuit.

Remark on asynchronous circuits

- The design of asynchronous circuits is more difficult in general because of timing problems involved. We will illustrate this next.

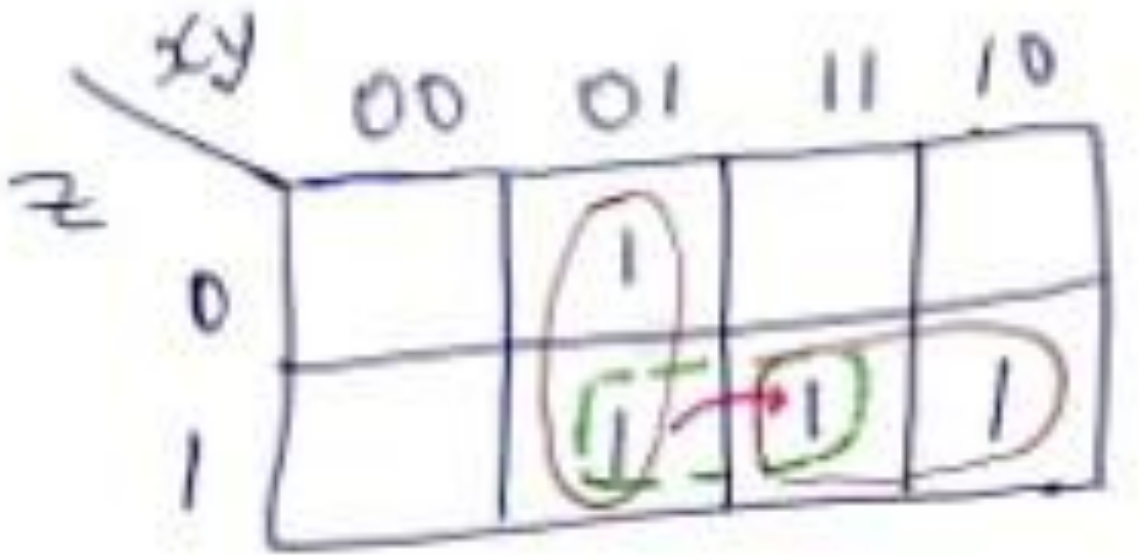
An issue with asynchronous circuit design

- There can be hazards (or glitches) which pose a challenge with asynchronous circuit design. Consider the implementation of the Boolean function

  $T(x,y,z) = \sum(2,3,5,7)$

  Its K-map is shown next.

K-map



Ignore the arrow for the moment

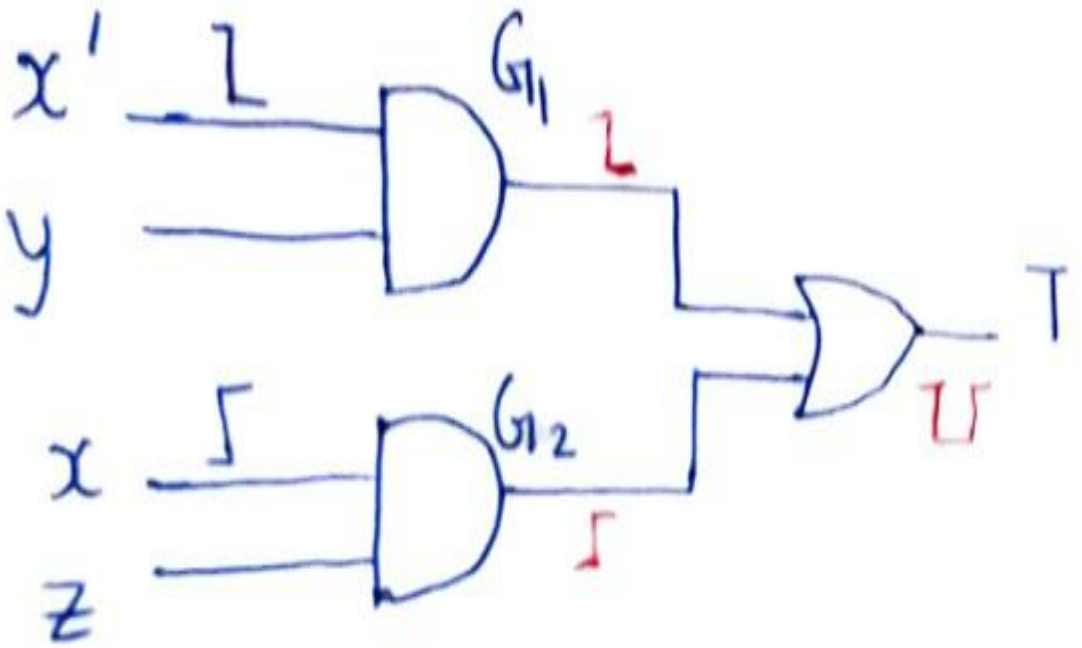
The function represented

- We note that the K-map leads to the simplified form given by

  T= x’y + xz

  Its realization using AND and OR gates is shown next.

Realization of T = x'y + xz



Signal transitions are indicated by ⌐ and ⌙

The hazard (or glitch) problem

- Suppose the value of inputs y and z is 1 and that the value of input x is changed from 0 to 1.
- As per the simplified expression for T, the value of T must remain at 1 irrespective of the change (in value of x).
- As the value of x changes, the transmission path through the network (shown in the previous slide) changes from gate G_1 to gate G_2.

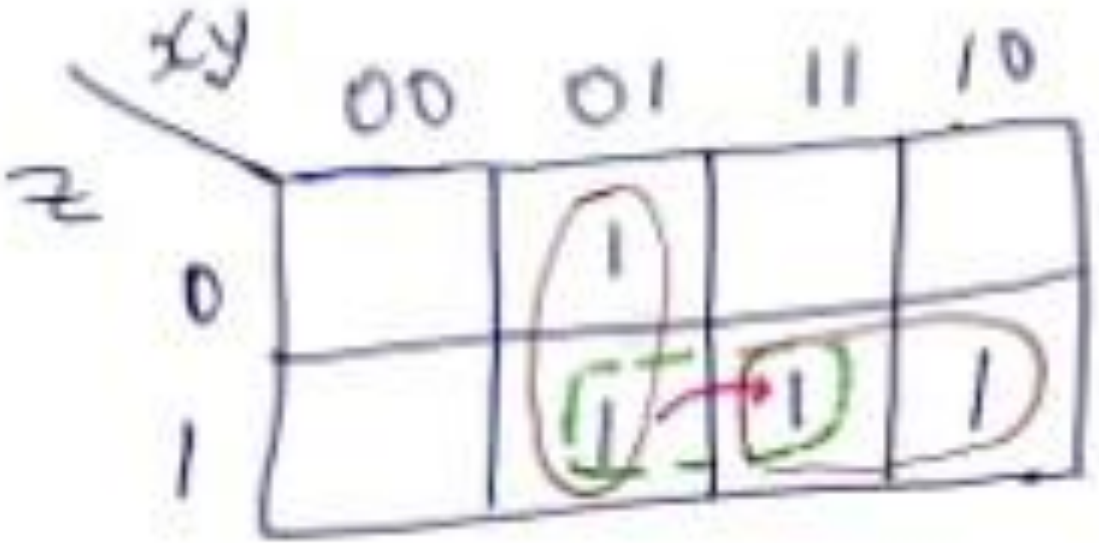
More on the hazard problem

- In an ideal situation, the change would be instantaneous and T will remain constant at 1.
- However, in practice, different delays are associated with gates G_1 and G_2 . As a result, there can be a short period when T is zero (if, for example, delay of G_1 is smaller than G_2 since the transmission x'y through G_1 will become 0 shortly before transmission xz through G_2).
- This phenomenon is called as a single-input-change static logic hazard. It can be misunderstood by another part of the circuit as a valid transition and cause incorrect behaviour.
- The single-input-change corresponds to one bit change.

How do we design to overcome the problem ?

- We need to ensure that the output T remains at 1.
- To this end, we can look at the K-map again.

Static logic hazard is indicated by the arrow in the K-map



Note x'yz and xyz are adjacent and the transition between adjacent input combinations that correspond to identical output values contains a single-input-change (SIC) static logic hazard.

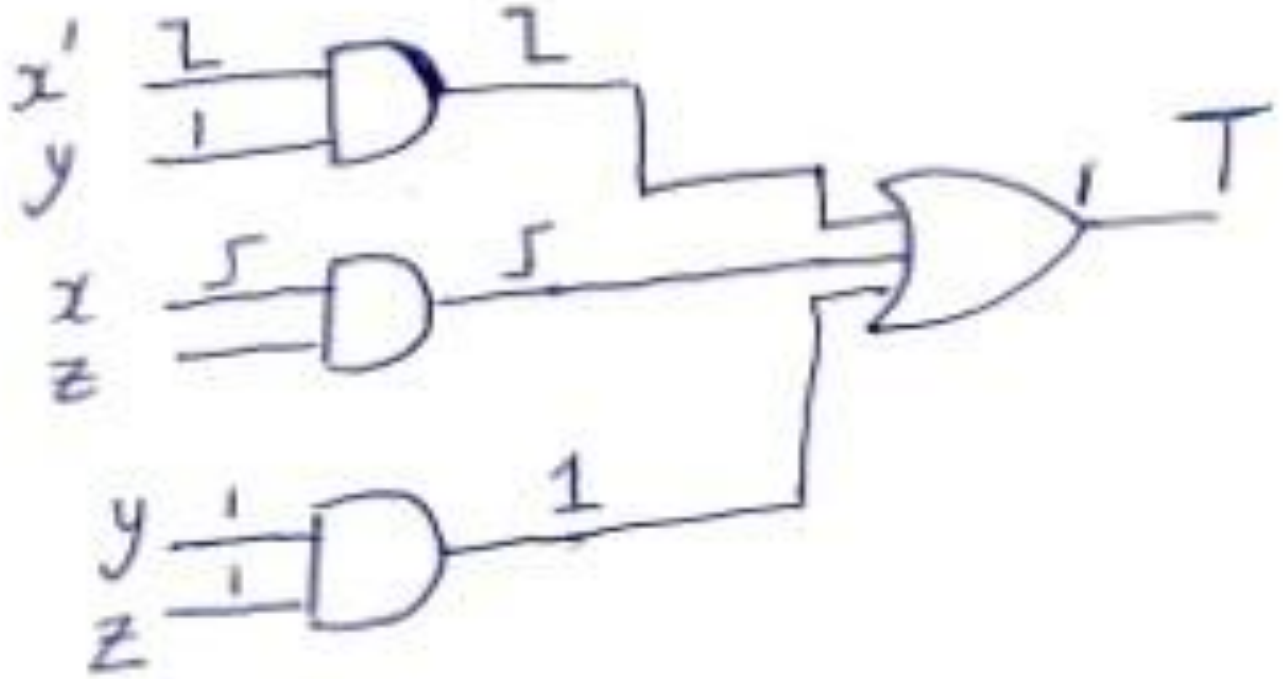
Include the prime implicant yz

- To remove the static logic hazard, we can include the prime implicant yz (corresponding to the dotted ellipse in the K-map) in the expression for T.
- So we have

  $T = x'y + xz + yz$

  The resulting circuit is shown next.

The modified hazard-free circuit



Explanation

- In the modified circuit, we have a third gate at the input side which has its inputs as y and z.
- When y = z = 1, note that T will be 1 regardless of the change at the input for the other two gates (which have x/x’ as an input).
- So regardless of the delays associated with x’ and x, the output T stays at 1.

References

- Z. Kohavi and N.K. Jha, Switching and Finite Automata Theory, Cambridge University Press, 2013.
- C. Roth, Fundamentals of Logic Design, West Publishing,circa 2002.